國立臺北科技大學
National Taipei University of Technology

資料結構
Data Structure

Lab 03

姓名: 夏宇柔
學號: 113840005

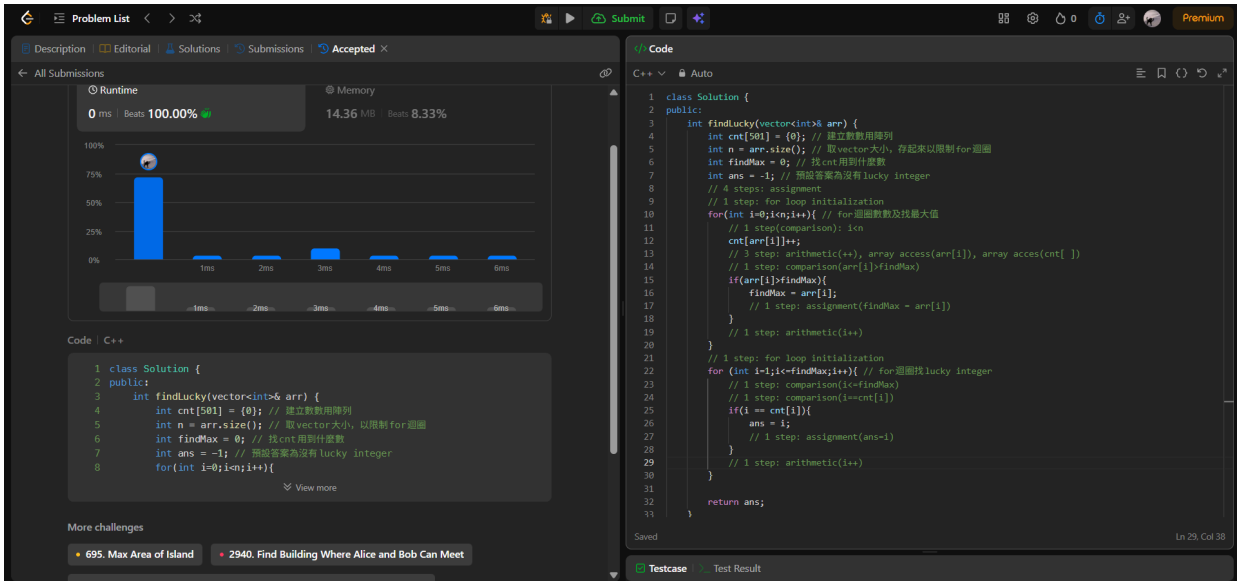
Lab03-Ex1.

Solve LeetCode 1394 and discuss time complexity.

Code

```cpp
class Solution {
public:
    int findLucky(vector<int>& arr) {
        int cnt[501] = {0}; // 建立數數用陣列
        int n = arr.size(); // 取vector大小，存起來以限制for迴圈
        int findMax = 0; // 找cnt用到什麼數
        int ans = -1; // 預設答案為沒有lucky integer
        // 4 steps: assignment
        // 1 step: for loop initialization
        for(int i=0;i<n;i++){ // for迴圈數數及找最大值
            // 1 step(comparison): i<n
            cnt[arr[i]]++;
            // 3 step: arithmetic(++), array access(arr[i]), array acces(cnt[ ])
            // 1 step: comparison(arr[i]>findMax)
            if(arr[i]>findMax){
                findMax = arr[i];
                // 1 step: assignment(findMax = arr[i])
            }
            // 1 step: arithmetic(i++)
        }
        // 1 step: for loop initialization
        for (int i=1;i<=findMax;i++){ // for迴圈找lucky integer
            // 1 step: comparison(i<=findMax)
            // 1 step: comparison(i==cnt[i])
            if(i == cnt[i]){
                ans = i;
                // 1 step: assignment(ans=i)
            }
            // 1 step: arithmetic(i++)
        }

        return ans;
    }
};
```



Discussion Section
The first loop iterates through the array once, taking O(n) time. The second loop runs up to the maximum value in the array, which is bounded by 500, so it is O(1). Therefore, the overall time complexity is O(n).